Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Planetary Systems

Mara Attia

Kapteyn Astronomical Institute
University of Groningen

2026

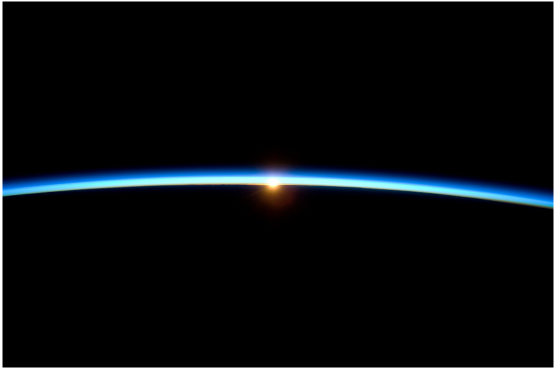
Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

- Classify atmospheres as primary, secondary or tertiary
- Derive the scale height from hydrostatic equilibrium and apply it across the solar system
- Explain the greenhouse effect and the escape mechanisms that decide which atmospheres survive

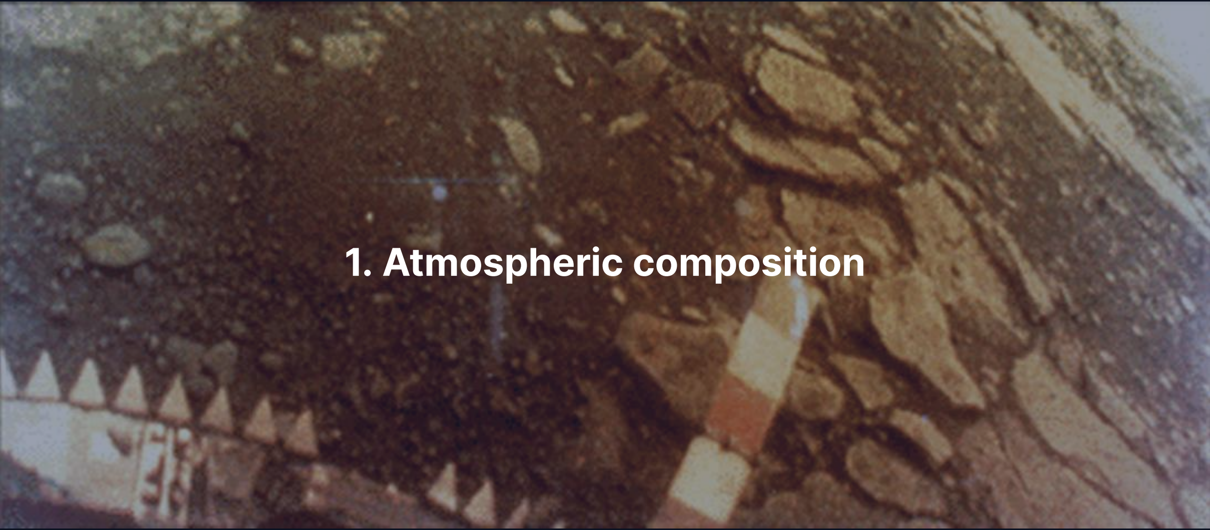
Recap: Lecture 4



Credit: NASA/ISS Expedition 21 crew, public domain

- Metal-silicate separation in magma oceans formed cores within ~30–60 Myr
- A dynamo needs a conducting fluid, convection and $\mathrm{Rm} \gtrsim$ 10–100
- Mars lost its dynamo between ~4.1 and 3.7 Ga, then its atmosphere

Today: The thin gaseous envelope that controls a planet's surface conditions.

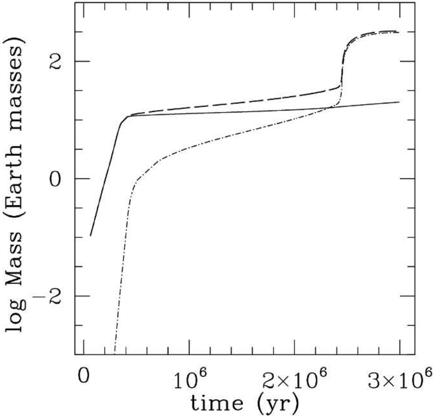


The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- **Gas giants** keep massive $\mathrm{H_2}$/He envelopes; **ice giants** only 10–20% of their mass
- Terrestrial planets are too small to retain primordial $\mathrm{H_2}$/He



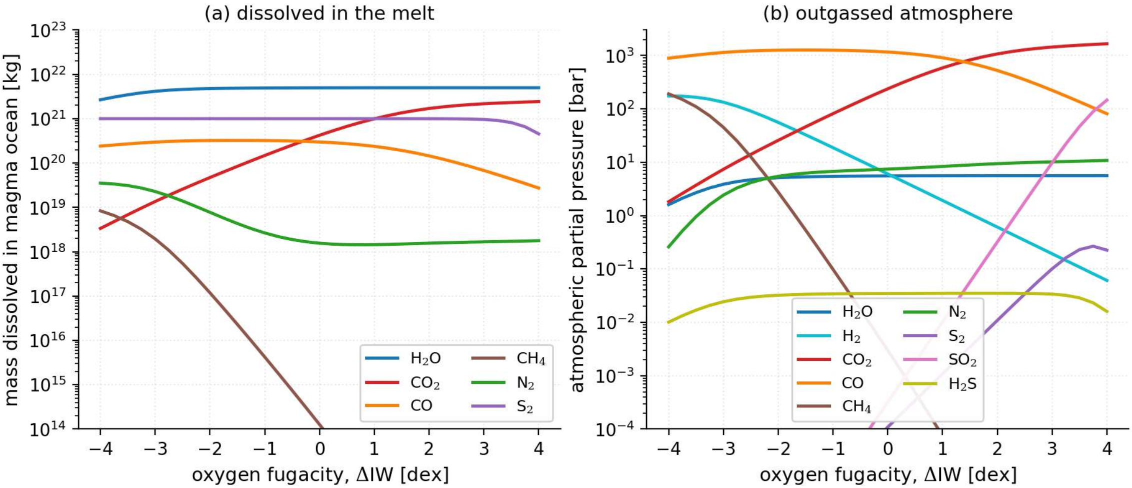
Growth of a giant planet by core accretion: runaway gas accretion builds the $\mathrm{H_2}$/He envelope once it outweighs the core. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

A primary atmosphere needs a core of ≳ 5–10 $M_\oplus$ before the disk disperses in ~3–10 Myr.

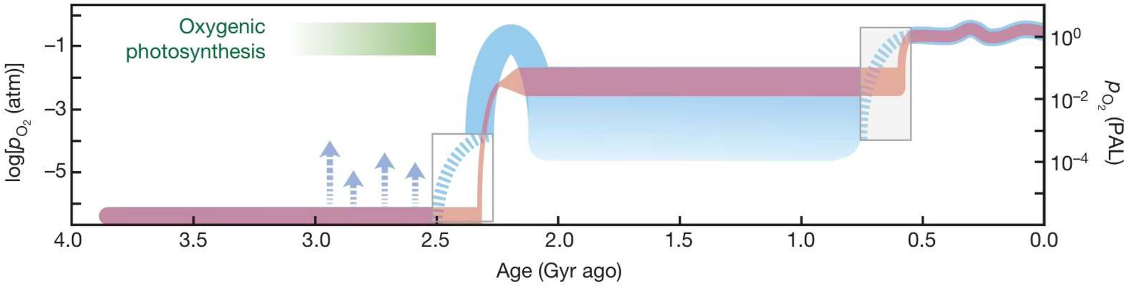
Secondary atmospheres

Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation follows the mantle **oxygen fugacity**: oxidising gives $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{N_2}$; reducing gives $\mathrm{H_2}$, CO, $\mathrm{N_2}$
- **Venus:** 96.5% $\mathrm{CO_2}$ at 92 bar; **Mars:** 95% $\mathrm{CO_2}$ at 6 mbar; **Titan:** $\mathrm{N_2}$ (from $\mathrm{NH_3}$) at 1.5 bar



Speciation of an outgassed atmosphere against melt oxygen fugacity: CO, H_2 and CH_4 when reducing, CO_2 and H_2O when oxidising. Course figure; model of Nicholls et al. (2024).



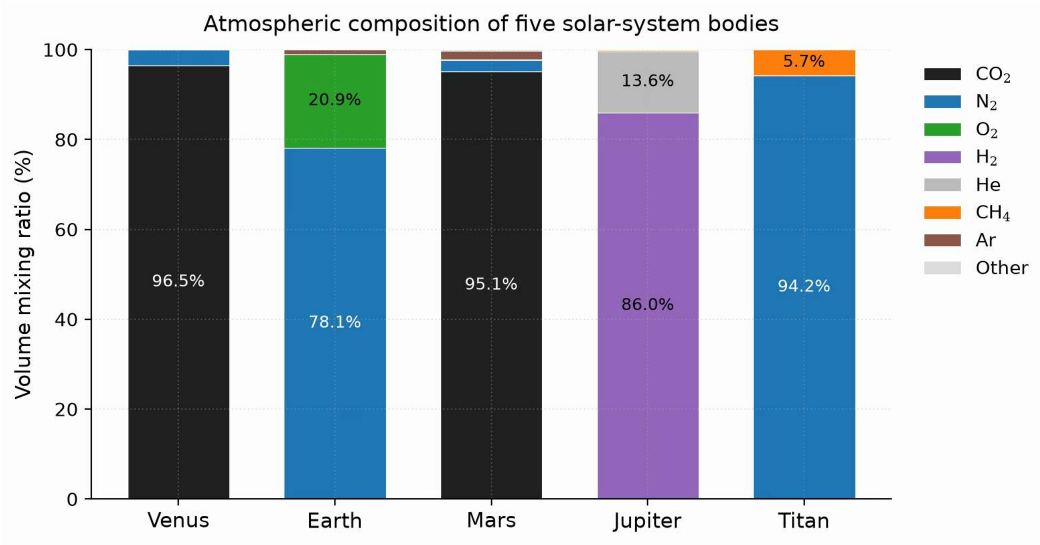
Earth's atmospheric oxygen through time: anoxic for ~2 Gyr, rising at the Great Oxidation Event near 2.4 Ga, modern levels in the Phanerozoic. Reproduced from Lyons et al. (2014).

Earth's atmosphere is **tertiary**: the outgassed $\mathrm{CO_2}$ and $\mathrm{N_2}$ inventory was Venus-like, oxygenic photosynthesis added the 21% $\mathrm{O_2}$, and that $\mathrm{O_2}$ is entirely biogenic and needs active life to persist.

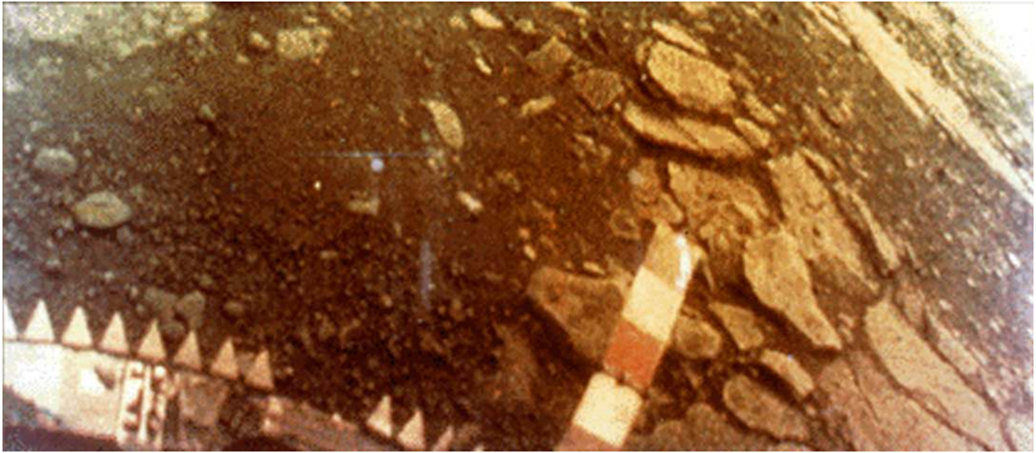
Comparative atmospheric properties

	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	CO_2	N_2	CO_2	H_2	N_2
Secondary gas	N_2	O_2	N_2	He	CH_4
μ	43.4	28.97	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

Surface pressures span four orders of magnitude, temperatures run from 94 to 737 K.



Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes. Course figure.



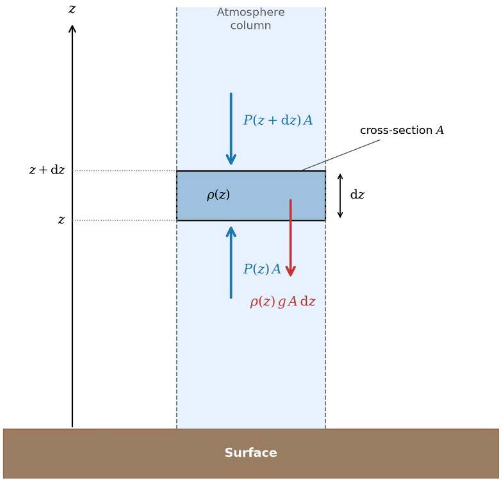
The surface of Venus seen by *Venera 13* on 1 March 1982. Credit: USSR/NASA NSSDC, public domain.

The lander survived 127 minutes at 92 bar and 737 K; the dense $\mathrm{CO_2}$ atmosphere and cloud deck filter sunlight to a dim orange glow.

2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

Hydrostatic equilibrium



Credit: Course figure

The net vertical force on a thin slab of area A and thickness $\mathrm{d}z$ vanishes:

$$P(z)A - P(z + \mathrm{d}z)A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Each layer supports the weight of all gas above it.

The ideal gas law and barometric formula

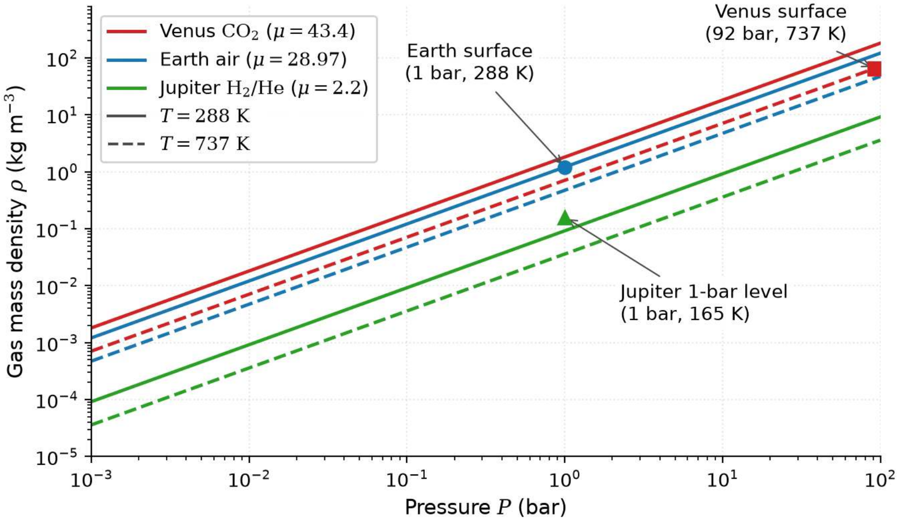
The ideal gas law $P = \rho\, k_{\mathrm{B}} T/(\mu\, m_u)$ in $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$ gives

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_{\mathrm{B}} T} P = -\frac{P}{H}$$

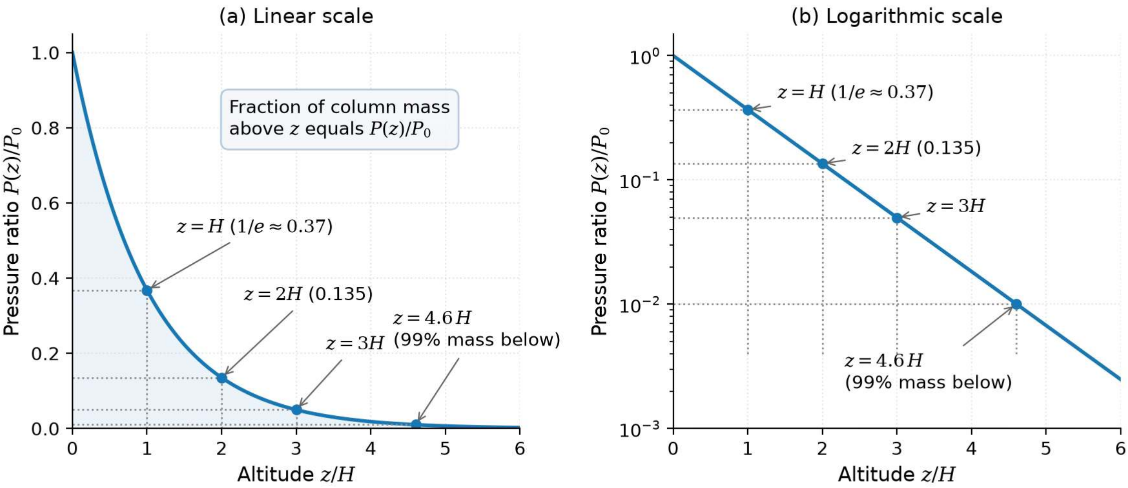
For an **isothermal** atmosphere (constant T):

$$\boxed{P(z) = P_0 \exp\left(-\frac{z}{H}\right)}$$

with $H = k_{\mathrm{B}} T/(\mu\, m_u\, g)$; the pressure drops by a factor of $e \approx 2.718$ each scale height.



Ideal gas law in atmospheric form, $\rho = P\mu m_u/(k_B T)$: density scales with pressure at fixed T, heavier or colder gas is denser. Course figure.

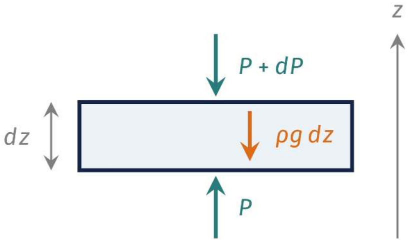


The barometric formula $P/P_0 = e^{-z/H}$: a factor e per scale height, 99% of the column mass below $4.6\,H$, a straight line on a logarithmic axis. Course figure.

3. Blackboard derivation: scale height

Sunrise over Earth’s limb from orbit. Credit: NASA

Goal and strategy



Goal: derive $H = k_B T / (\mu\, m_u\, g)$ from hydrostatic equilibrium and the ideal gas law.

- Balance the forces on a thin slab
- Replace ρ by P with the ideal gas law
- Read off the length scale, integrate, compute

One length, H, sets how fast pressure falls with height.

Deriving the scale height

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_{\mathrm{B}} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\, \mu\, m_u}{k_{\mathrm{B}} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_{\mathrm{B}} T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}}$$

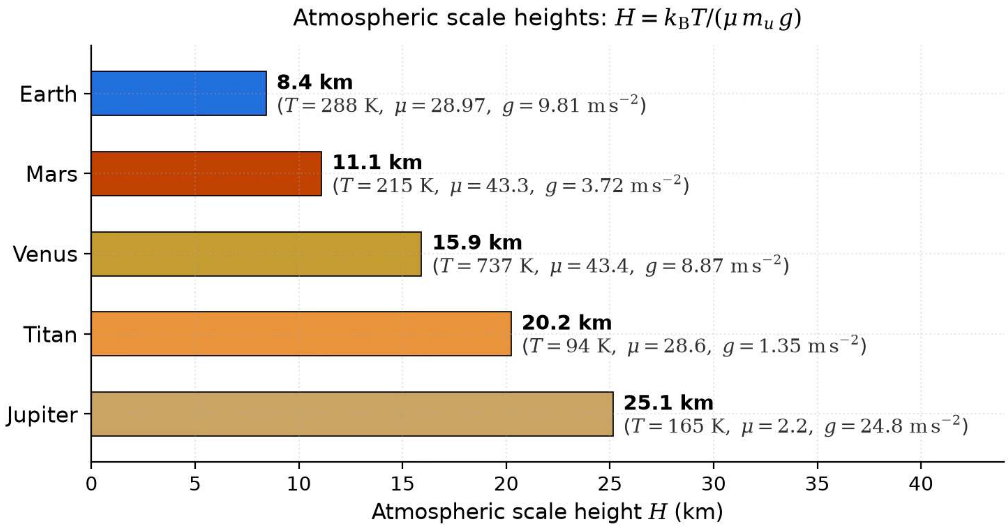
- **Higher** T → larger H: hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H: harder to loft
- **Stronger gravity** g → smaller H: more compressed

H balances thermal energy against gravitational binding; integrated over the column, the surface pressure is the column weight, $P_0 = m_{\mathrm{col}}\, g$.

Scale heights across the solar system

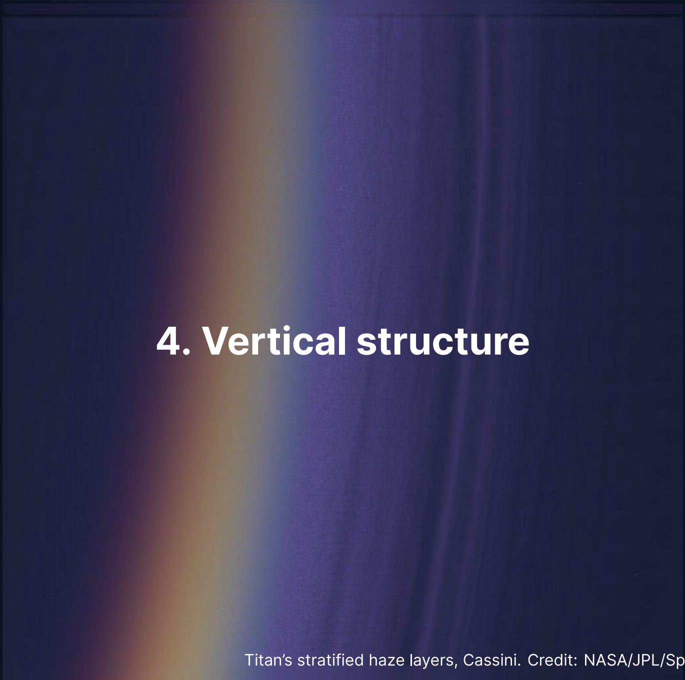
Body	T **(K)**	μ	g **(m s^{-2})**	H **(km)**
Venus	737	43.4	8.87	15.9
Earth	288	28.97	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

- Jupiter: large H despite strong gravity, because H_2 is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot, but heavy molecules and decent gravity yield a moderate H

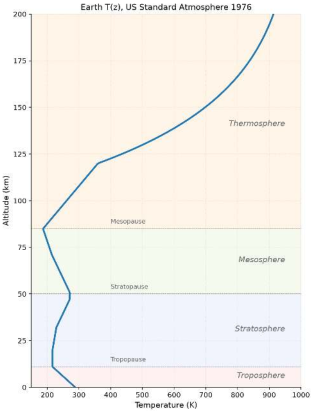


Scale heights recomputed from T, μ and g: light gas (Jupiter) and weak gravity (Titan) give the largest H, Earth the smallest. Course figure.

4. Vertical structure



Titan’s stratified haze layers, Cassini. Credit: NASA/JPL/Space Science Institute (PIA06160)



Earth's atmospheric layers, defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$. Course figure; US Standard Atmosphere 1976.

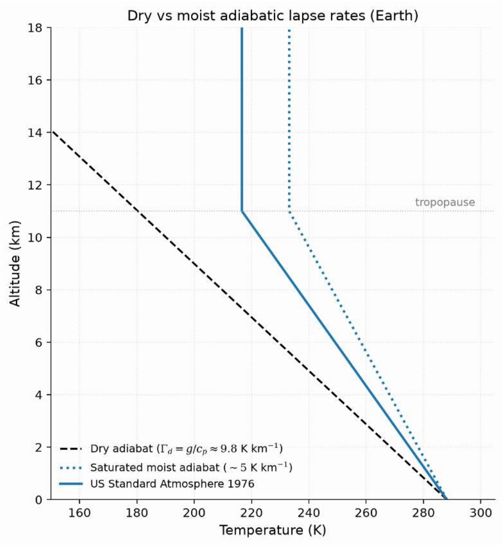
Troposphere (0–12 km): T falls, convective; **stratosphere** (12–50 km): T rises by UV ozone heating; **mesosphere** (50–85 km): T falls to the ~190 K mesopause; **thermosphere** (>85 km): EUV heating, T > 1000 K.

The adiabatic lapse rate

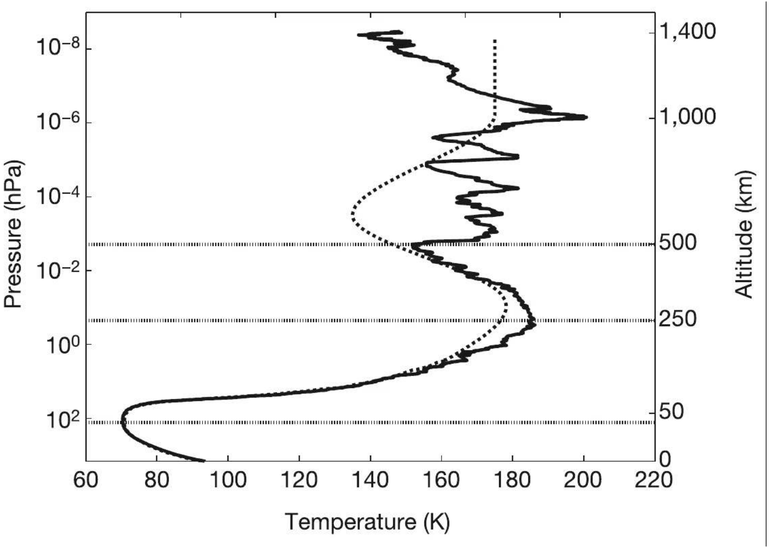
In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

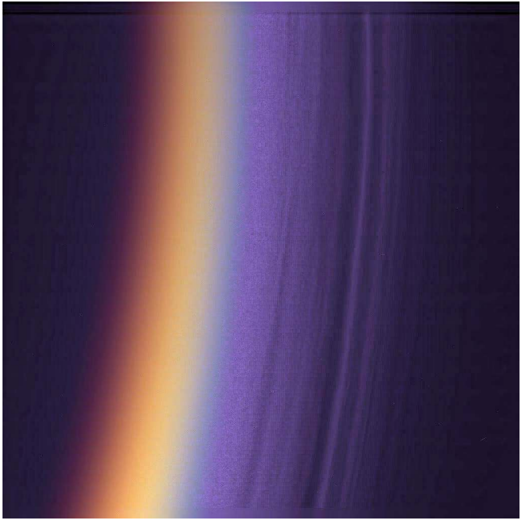
- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8\ \mathrm{K\,km^{-1}}$; observed ~6.5 $\mathrm{K\,km^{-1}}$ because of latent heat
- Rising air expands, does work on its surroundings, and cools
- **Tropopause height:** ~8 km (poles) to ~17 km (equator)



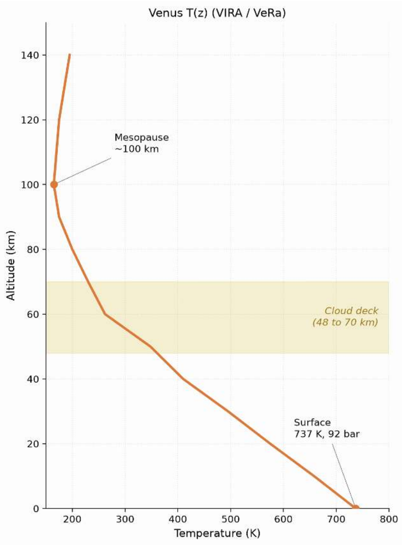
Dry and moist adiabats against the US Standard Atmosphere 1976: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.



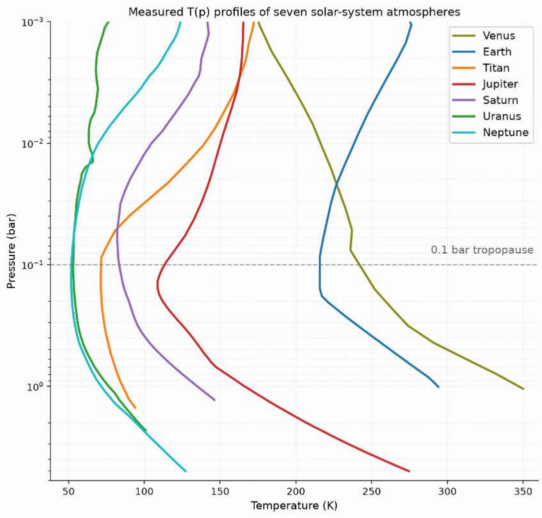
Titan's temperature profile from the Huygens HASI descent: the 94 K surface, the 0.1 bar tropopause and the stratosphere measured in situ. Fulchignoni et al. (2005).



Detached haze layers in Titan's upper atmosphere, Cassini ISS: photochemical haze up to 500 km altitude absorbs solar UV and heats the stratosphere. Credit: NASA/JPL/Space Science Institute, public domain.



Venus temperature profile from Pioneer Venus/VIRA and Venus Express: without ozone, T falls monotonically from 737 K at the surface to the mesopause near 100 km. Course figure.



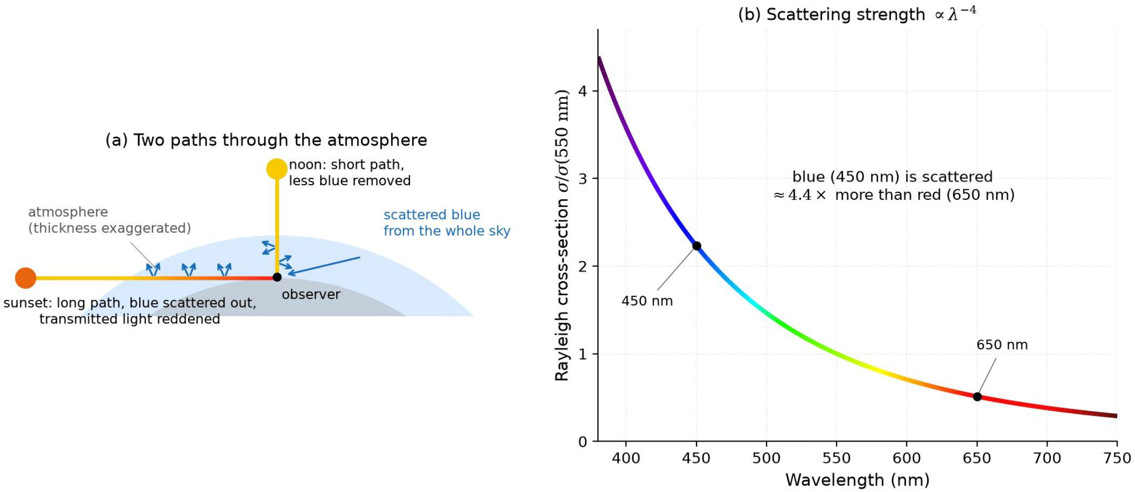
Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres turn from convective to radiative near 0.1 bar, a universal tropopause. Digitized from Robinson & Catling (2014), Fig. 1.

5. Radiative transfer basics

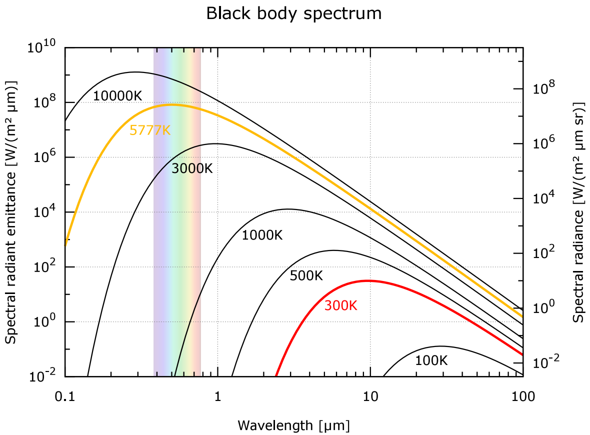
Earth’s thin atmosphere seen from orbit. Credit: NASA/ISS

Three interactions: radiation and matter

- **Absorption:** photon energy becomes internal molecular energy; key absorbers $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{CH_4}$, $\mathrm{O_3}$, $\mathrm{N_2O}$
- **Emission:** warm gas radiates at the absorbed wavelengths (Kirchhoff's law); only emission from $\tau \lesssim 1$ reaches space
- **Scattering:** photons redirected without absorption; Rayleigh ($\propto \lambda^{-4}$) and Mie scattering



Rayleigh scattering and the colour of the sky: short paths at noon keep the blue scattered light, long slant paths at sunset transmit only red; the cross-section scales as λ^{-4}. Course figure.



Blackbody spectra of the Sun (~5800 K, peak ~0.5 μm) and a planet (~300 K, peak ~10 μm).
Credit: Wikimedia, public domain.

This **wavelength separation** is the basis of the greenhouse effect: an atmosphere can be transparent in the visible but opaque in the infrared.

Optical depth and the Beer-Lambert law

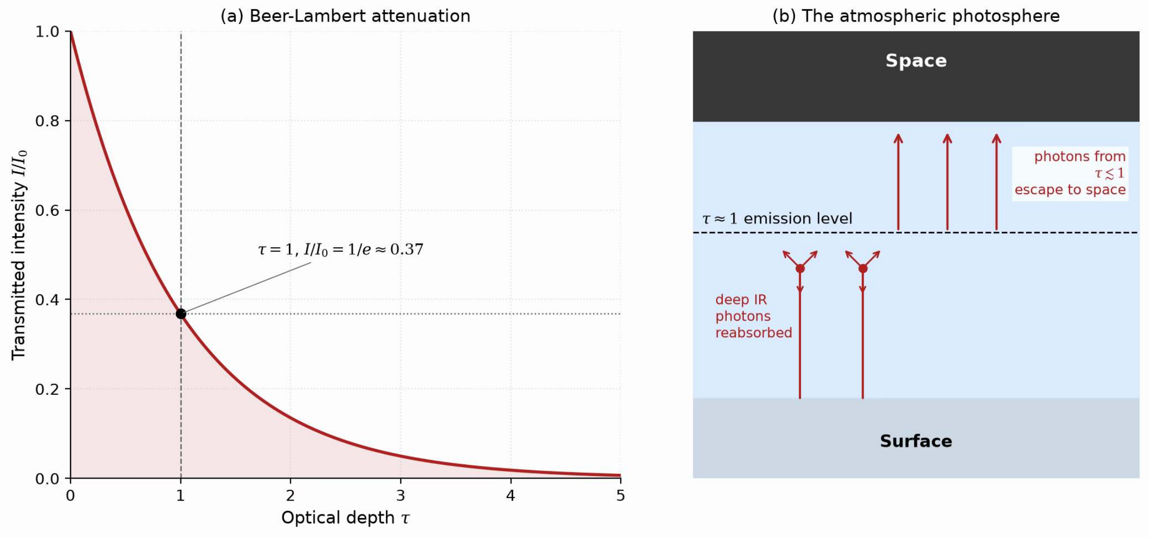
Optical depth τ measures the cumulative absorption along a path:

$$\tau = \int_0^s \kappa \, \rho \, \mathrm{d}s'$$

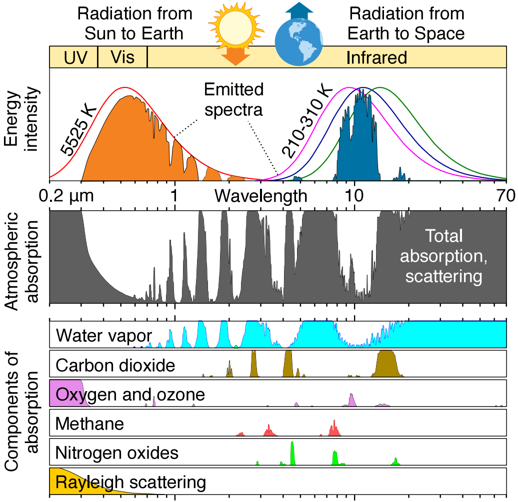
Beam intensity decays following the **Beer-Lambert law**:

$$\boxed{I = I_0 \, e^{-\tau}}$$

- **Optically thin** ($\tau \ll 1$): radiation passes freely through the gas
- **Optically thick** ($\tau \gg 1$): photons are absorbed and re-emitted many times
- **Photosphere** ($\tau \approx 1$): thermal radiation escapes directly to space



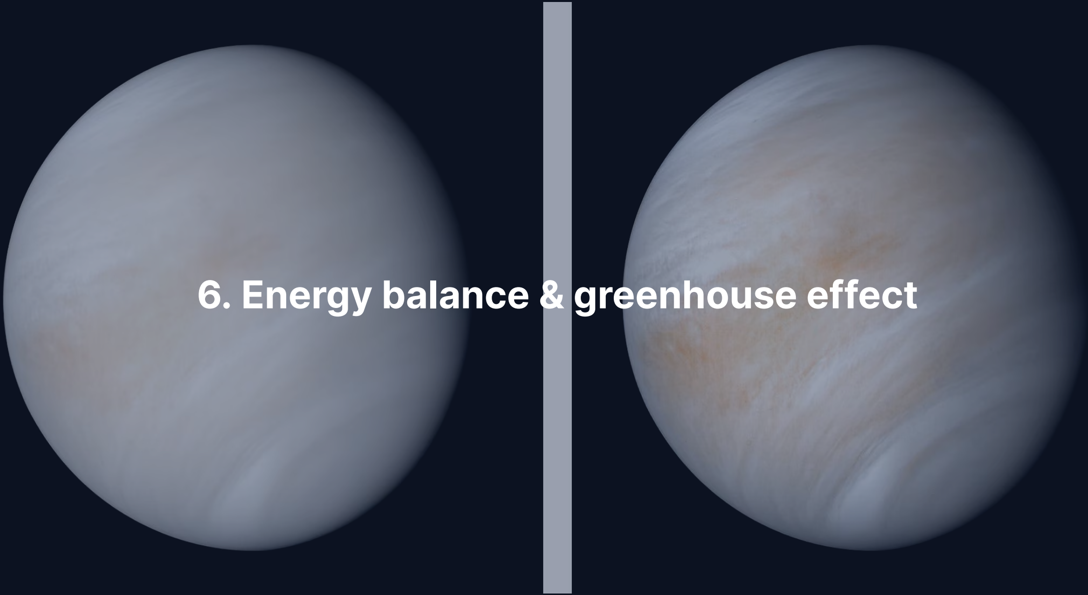
The atmospheric photosphere: thermal radiation escapes to space from the $\tau \approx 1$ level, which sets the effective temperature. Course figure.



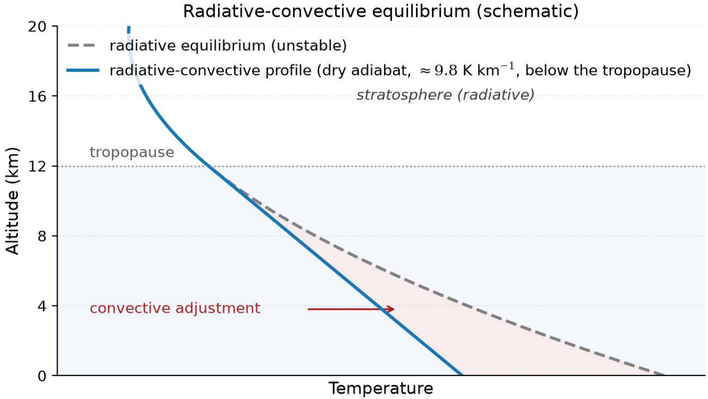
Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb the outgoing infrared while visible sunlight and the 8–13 μm window pass. Credit: Wikimedia Commons, public domain.

Earthrise, Apollo 8 (1968). Credit: NASA

6. Energy balance & greenhouse effect



Venus in ultraviolet light. Credit: NASA/JPL-Caltech



Radiative-convective equilibrium (schematic): the radiative profile is superadiabatic near the surface, convection holds the troposphere on the dry adiabat. Course figure.

Greenhouse opacity makes the **troposphere** convective, the optically thin **stratosphere** sits in radiative equilibrium, and the **tropopause** is where the radiative gradient becomes shallower than the adiabat.

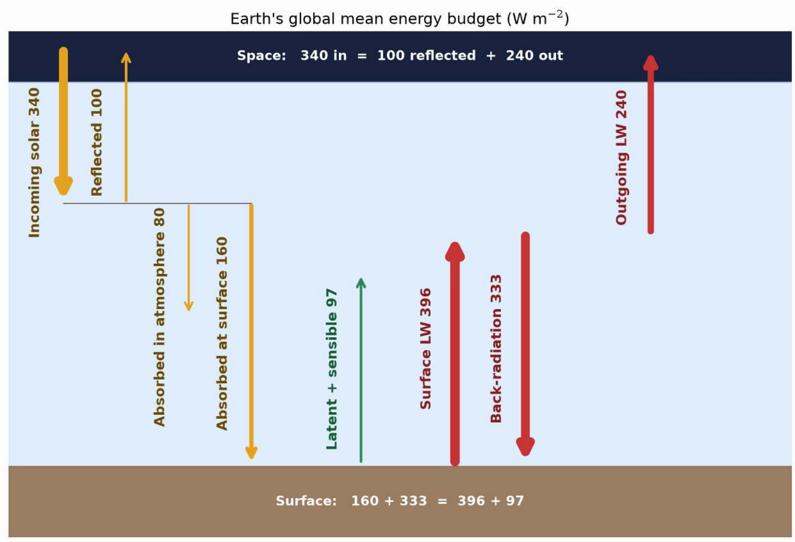
Planetary energy balance

Absorbed stellar power equals emitted thermal radiation:

$$(1 - A)\frac{S}{4} = \sigma T_{\mathrm{eff}}^4 \qquad \text{where} \quad S = \frac{L_\star}{4\pi d^2}$$

- A: Bond albedo; S: stellar flux ($S_0 = 1361\ \mathrm{W\,m^{-2}}$ at Earth)
- Factor $1/4$: intercepting cross-section πR_p^2 over the emitting sphere $4\pi R_p^2$

$$\boxed{T_{\mathrm{eff}} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\,d^2}\right]^{1/4} = \left[\frac{(1 - A)\,S}{4\sigma}\right]^{1/4}}$$



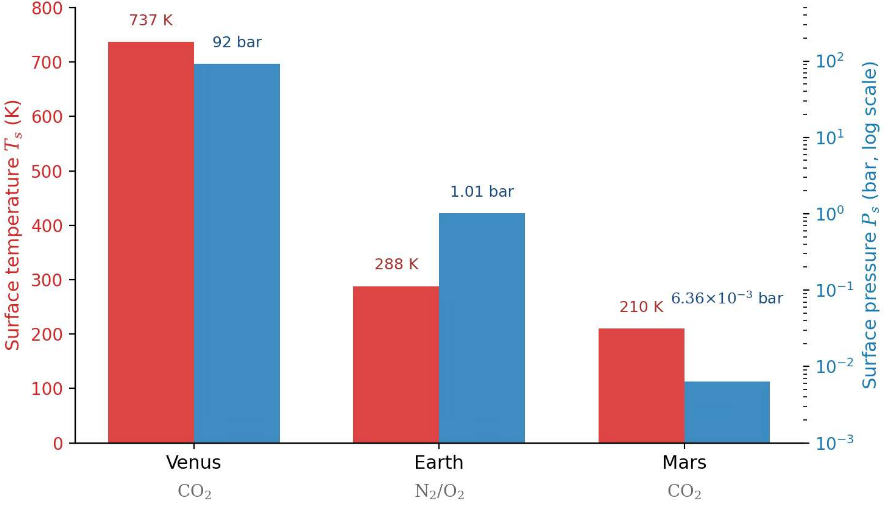
Earth’s globally averaged energy budget in $W\,m^{-2}$: back-radiation of 333 $W\,m^{-2}$ supplies more energy to the surface than direct sunlight (160 $W\,m^{-2}$). After Trenberth et al. (2009).

Effective vs. surface temperature

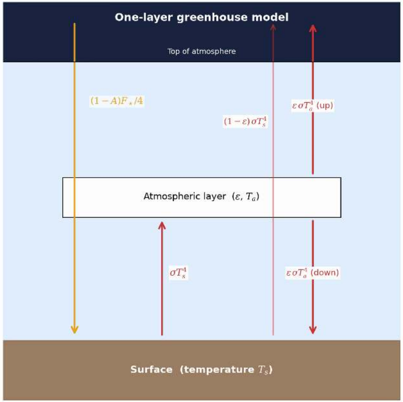
Body	A	d **(AU)**	T_{eff} **(K)**	T_{surf} **(K)**	ΔT **(K)**
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\mathrm{surf}} - T_{\mathrm{eff}}$ is the **greenhouse effect**: 510 K on Venus, 33 K on Earth, ~5 K on Mars (6 mbar, no water-vapour amplifier).



Surface temperature and pressure of Venus, Earth and Mars: similar bulk compositions span a factor 10^4 in pressure and 500 K in temperature through greenhouse warming. Course figure; data from NASA.



The one-layer greenhouse model: sunlight in, infrared absorbed and re-emitted half up, half down. Course figure.

Sunlight (~0.5 μm) reaches the surface, the surface emits thermal IR (~10–15 μm), greenhouse gases absorb it and re-emit half downward, and that extra energy makes $T_{\rm surf} > T_{\rm eff}$.

One-layer greenhouse model

A single layer with infrared emissivity ε, transparent to visible sunlight:

Flux balance:

Atmosphere: $\varepsilon\sigma T_s^4 = 2\varepsilon\sigma T_a^4$

Surface: $(1 - A)\frac{F_\star}{4} + \varepsilon\sigma T_a^4 = \sigma T_s^4$

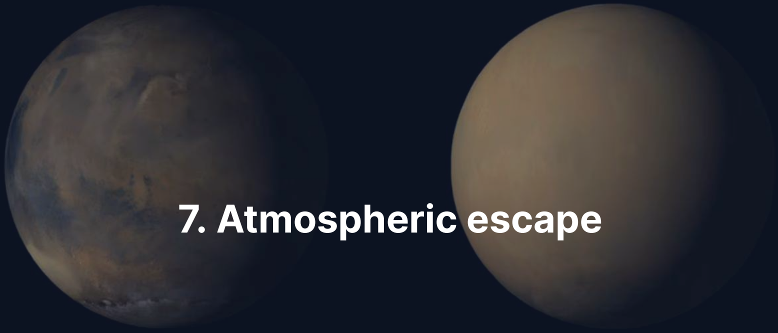
TOA: $(1 - A)\frac{F_\star}{4} = \sigma T_{\mathrm{eff}}^4$

- $\varepsilon = 0$ (transparent): $T_s = T_{\mathrm{eff}}$
- $\varepsilon = 1$ (opaque): $T_s = 2^{1/4}\, T_{\mathrm{eff}} \approx 1.19\, T_{\mathrm{eff}}$ (Earth: ≈303 K)

Surface temperature:

$$\boxed{T_s = T_{\mathrm{eff}}\left(\frac{2}{2 - \varepsilon}\right)^{1/4}}$$

7. Atmospheric escape



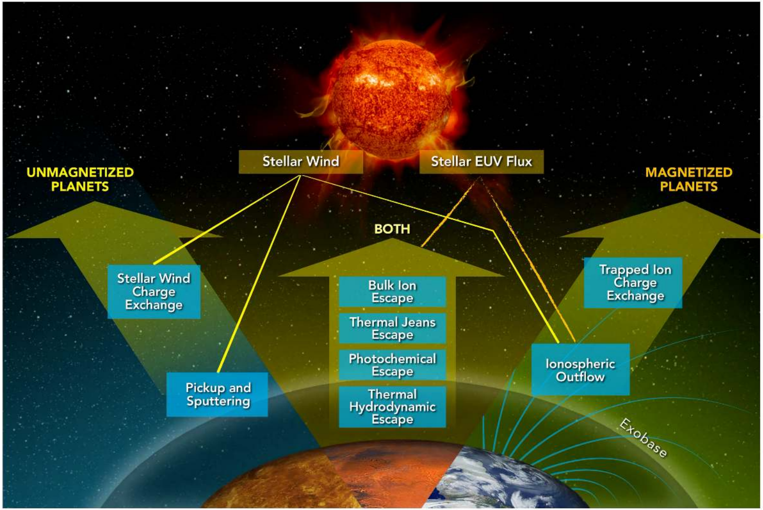
The 2018 global dust storm on Mars. Credit: NASA/JPL-Caltech/MSSS

Thermal (Jeans) escape

The high-velocity tail of the Maxwell-Boltzmann distribution can exceed the escape speed. The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{G M_p m}{k_B T_{exo} r_{exo}} = \frac{v_{esc}^2}{v_{th}^2}, \qquad v_{th} = \sqrt{2 k_B T_{exo} / m}$$

- $\lambda_J \gg 1$: escape is negligible because very few molecules exceed v_{esc}
- $\lambda_J \lesssim 2$–3: thermal escape drives rapid atmospheric loss
- Escape flux $\Phi_J \propto (1 + \lambda_J) e^{-\lambda_J}$: exponential in λ_J



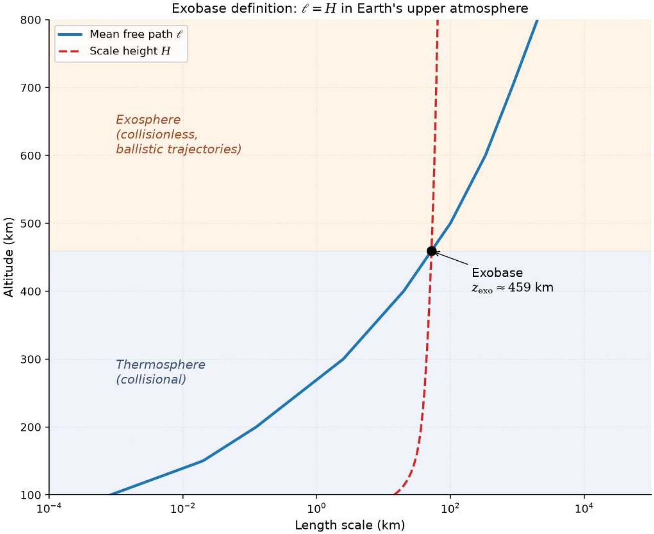
The main atmospheric escape processes: thermal, photochemical and ion channels act on all planets, and the magnetic field selects which dominate. Gronoff et al. (2020).

Jeans parameter for Earth and Mars

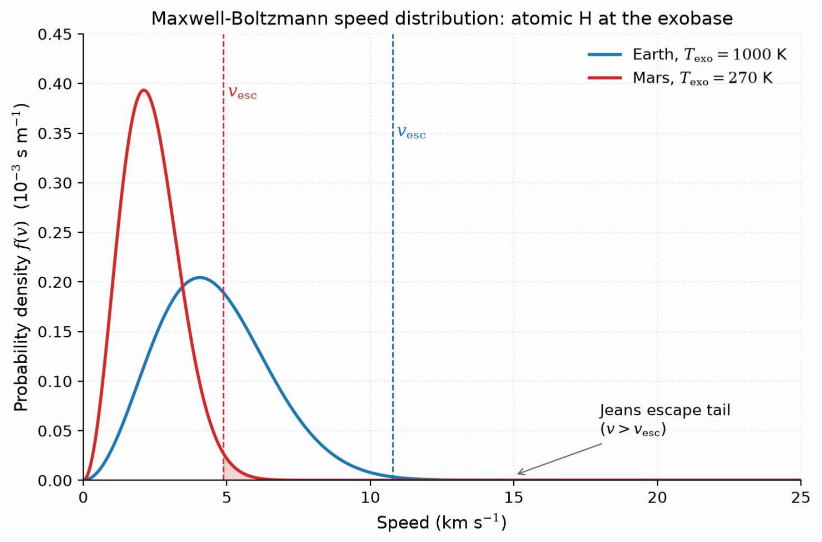
Exobase temperatures: 1000 K (Earth), 270 K (Mars).

Species	m **(u)**	λ_J **(Earth)**	λ_J **(Mars)**
H	1	7.0	5.3
H_2	2	14	11
He	4	28	21
N_2	28	200	150
CO_2	44	310	230

$\lambda_J \propto m$: atomic H escapes steadily from both planets, while N_2 and CO_2 stay gravitationally bound.



The exobase: the altitude where the mean free path ℓ equals the scale height H, collisional fluid below and ballistic flight above. Course figure.



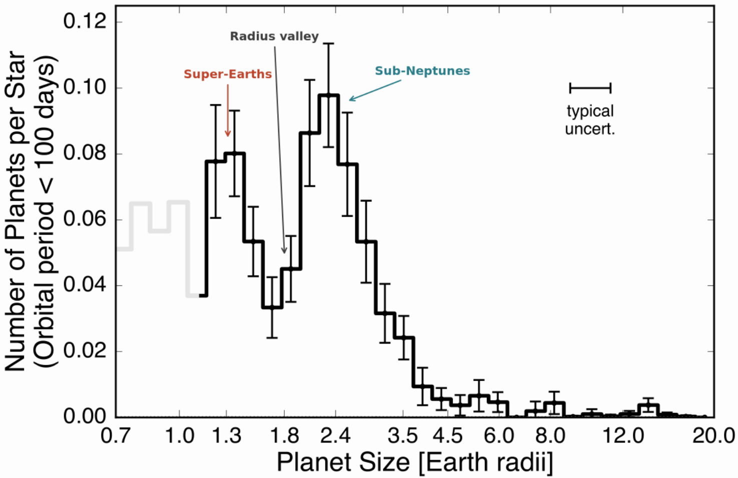
Maxwell-Boltzmann speed distributions for atomic H at the Earth and Mars exobases: the escaping fraction is the tail beyond $v_{\rm esc}$, scaling as $e^{-\lambda_J}$. Course figure.

Hydrodynamic escape

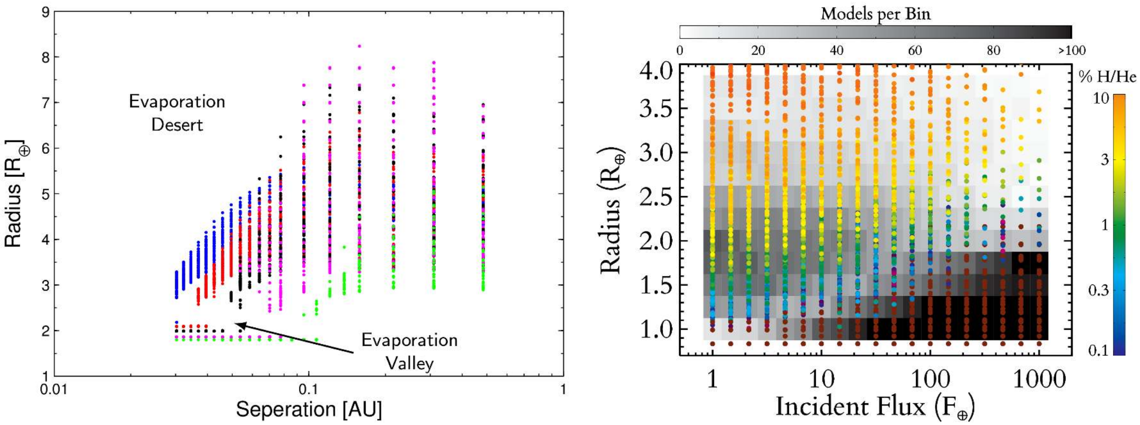
Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Strongest in the first few hundred Myr, when stellar EUV is 10–100× higher
- Strips $\mathrm{H_2}$-rich primary envelopes from planets up to several $M_\oplus$; outflowing H drags heavier volatiles along
- The primary mechanism behind the exoplanet radius valley at ~1.5–2 $R_\oplus$

Source: Hunten et al. (1987); Owen (2019)



The radius valley in the California-Kepler Survey: a deficit near 1.8 $R_\oplus$ separates stripped rocky cores from envelope-retaining sub-Neptunes. After Fulton et al. (2017).



Population synthesis after 10 Gyr of EUV-driven escape: an empty evaporation desert and a valley near 1.5 to 2 $R_{\oplus}$. After Owen & Wu (2013), Lopez & Fortney (2013) and Owen (2019).

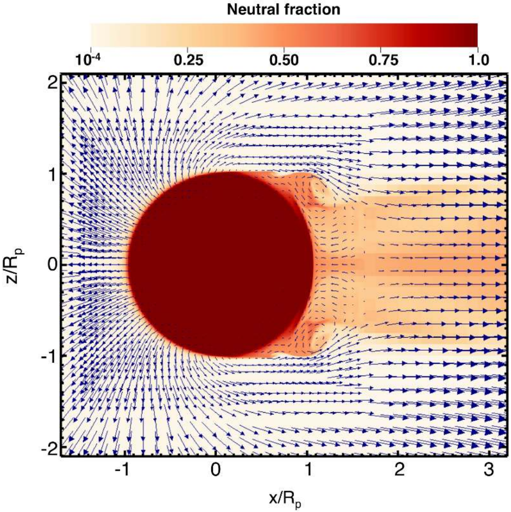
Energy-limited escape rate

Stellar EUV absorbed in the upper atmosphere drives hydrodynamic mass loss:

$$\dot{M}_{\mathrm{EL}} = \eta \, \frac{\pi \, F_{\mathrm{EUV}} \, R_p^3}{G \, M_p}$$

- Efficiency $\eta \approx 0.1$–0.2 converts incident EUV flux into expansion work
- Scales as R_p^3: extended, low-density atmospheres lose mass fastest
- Scales inversely with M_p: low gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)

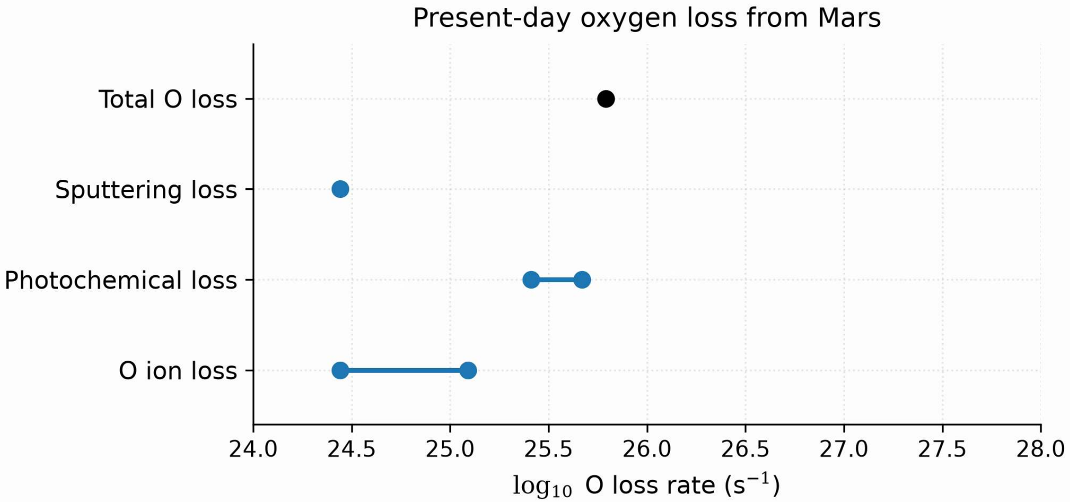


Hydrodynamic outflow from a hot Jupiter under stellar EUV heating: the gas accelerates outward into a night-side wake with Kelvin-Helmholtz instabilities. Reproduced from Owen (2019), Fig. 2; simulation from Tripathi et al. (2015).

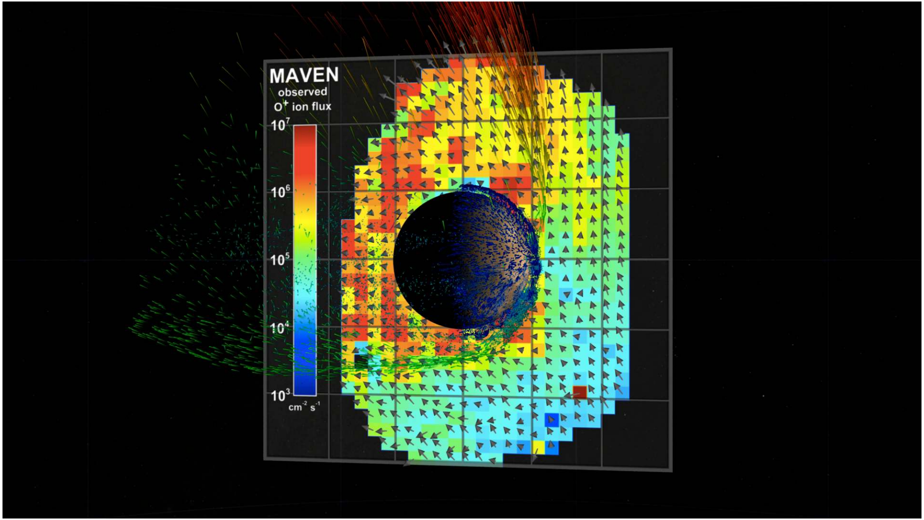
Non-thermal escape mechanisms

- **Sputtering** and **ion pickup:** solar wind ions eject neutrals, and photoions are swept away by the interplanetary field
- **Photochemical:** photodissociation yields suprathermal atoms above v_{esc}
- **Impact erosion:** giant impacts shock and eject atmospheric columns

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.



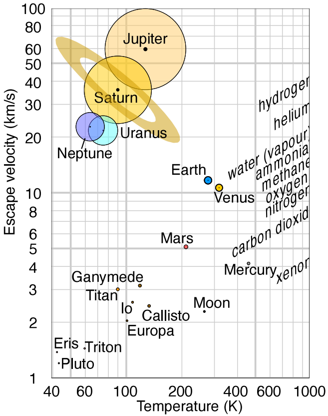
Present-day oxygen loss from Mars by channel: photochemical escape dominates, and with hydrogen escape the total reaches the 2–3 kg s^{-1} MAVEN measures. After Jakosky et al. (2018).



The ion plume of Mars from MAVEN: energetic ions (red) leave in a plume, the bulk (green) along the tail, with no global field to keep the solar wind out. Credit: NASA Scientific Visualization Studio (SVS 4370), public domain.

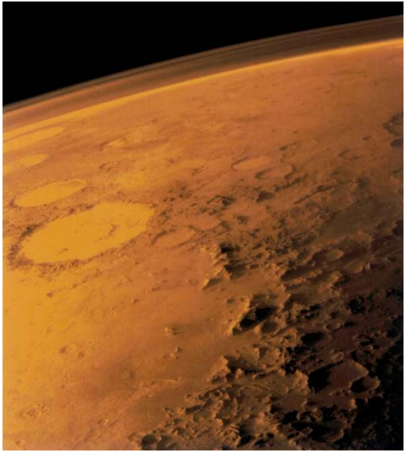
8. Atmospheric retention

Jupiter’s Great Red Spot. Credit: NASA/JPL-Caltech/SwRI/MSSS



Escape velocity against temperature, with the retention lines for each gas. Credit: Wikimedia, CC BY-SA 4.0.

Long-term retention needs $v_{\mathrm{esc}} \gtrsim 6\, v_{\mathrm{th}}$: gas giants keep every species, Earth and Venus lose the light gases H and He, Mars is marginal, cold Titan keeps N_2, the Moon and Mercury keep nothing.

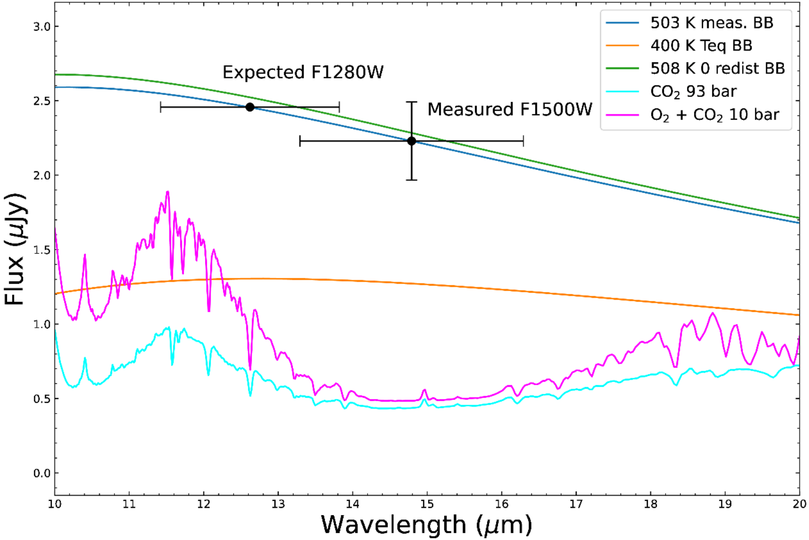


The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL.

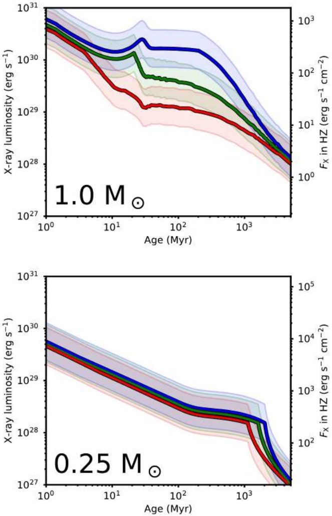
After the dynamo died near ~4.1 Ga the solar wind eroded a dense early atmosphere, recorded by valley networks and lakes, down to today’s 6 mbar: no stable liquid water, little greenhouse warming.

9. Recent advances & outlook

Earth’s atmosphere. Credit: NASA/ISS



JWST/MIRI 15 μm thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).



X-ray luminosity evolution of Sun-like stars and M dwarfs: low-mass stars stay saturated for over 1 Gyr and strip close-in rocky planets for longer. Reproduced from Johnstone et al. (2021), Fig. 11.

Summary

1. **Atmosphere types**: primary (captured), secondary (outgassed), tertiary (biogenic); four orders of magnitude in surface pressure
2. **Hydrostatic balance** and the **scale height** $H = k_{\mathrm{B}}T/(\mu m_u g)$ give $P = P_0 e^{-z/H}$; convection sets $\Gamma_d = g/c_p$ below the 0.1 bar tropopause
3. **Greenhouse warming** from $\tau \gtrsim 1$ raises T_s above T_{eff}; **thermal escape** is negligible when $v_{\mathrm{esc}} \gtrsim 6v_{\mathrm{th}}$; hydrodynamic and non-thermal loss follow other rules

Next: Lecture 6

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`